

Table 13.5: Allocation to Irrigation Projects as percentage to GDP

Year	1981-82	1990-91	1995-96	2000-01	2005-06	2011-12	2013-14
Share	1.4	0.7	0.7	0.7	0.8	0.6	0.5

Source: Shantanu De Roy(2017), “Economic Reforms and Agricultural Growth in India”, Economic and Political Weekly, vol LII no 9, March 4, 2017, p.70 (Table 6).

The low irrigation cover for various crops has led to severe rainfall dependency. One finds a high correlation between production and rainfall for pulses and oilseeds.

13.3.3 Failure to Evolve and Adopt New Technologies

India was able to avail of the potential of seed-fertilizer technology because of favourable international research collaborations. However, the country has failed to make a major breakthrough in frontal areas like biotechnological research. In India, public funding for agriculture Research and Education (R&E) is contributed by both Centre and States with around 55.4 per cent of the total allocation contributed by the Centre and 44.6 per cent by states. Lack of investment in research and technology in agriculture has resulted in the non-availability of any new cost– reducing technology in agriculture and has led to declining input efficiency. One of the main reasons for the low levels of yields in Indian agriculture has been the unsatisfactory spread of new technological practices including HYVs of seeds and usage of fertilizers and pesticides and inadequate spread of farm management techniques and other practices such as soil conservation and crop rotation.

Between 2000-01 and 2014-15, India’s total R & E expenditure for agriculture and allied activities in real terms (2004-05 prices) has increased from Rs. 31.1 billion to Rs. 61.6 billion (a compound annual growth rate of 5 per cent). However, as a percentage of GDP from agriculture (GDPA), it amount to about 0.54 per cent in 2014-15.

13.3.4 Shrinking Farm Size

The shrinking size of average land holding of an Indian farmer has held back agricultural productivity and there is not much that can be done about this. Due to smaller holding size majority of Indian farms find it difficult to access institutional credit, access new technology, adopt more efficient form of farm practices and undertake more capital-intensive investment for land improvement. The average landholding size of a household has shrunk marginally to 1.1 hectare (ha) in 2015-16 from 1.16 ha three years ago, according to a rural survey carried out by the National Bank for Agriculture and Rural Development (NABARD). The said survey shows that nearly one-thirds of farmers have land parcels smaller than 1 ha, while 37 per cent of farm households owned land parcels of smaller than 0.4 ha, another 30 per cent had holdings which fall between 0.41 and 1.0 ha. Only 16 per cent agricultural households owned landholdings in between 1 to 2 ha and only 13 per cent of farmers have holdings bigger than 2 ha.

Table 13.6: Trends in Farm Size Holdings in India (Unit in Million)

Holding	1970-71	1980-81	1990-91	2000-01	2010-11
Marginal	36	50	83	75	93
Small	13	16	20	23	25
Medium	19	21	22	21	20
Large	3	2	2	1	1

Note: Marginal: up to 1 hectare, Small: 1-2 hectares, Medium: 2-10 hectares, Large: over 10 hectares. Sources: Agriculture Census 2011

13.3.5 Non-availability or Inadequate Credit from Formal Sources

Shantanu De Roy (2017) using the data from RBI's Handbook of Statistics for Indian Economy computed the trends in (a) opening of new rural branches, (b) credit-deposit ratio in rural areas and (c) shares of priority sector and agriculture in total outstanding credit of Commercial Banks which are presented in Figures (Figure 13.2A to 13.2C). It may be observed that reduction in bank branches in rural areas and declining credit-deposit ratios led to increased dependence of smaller cultivators on private moneylenders at exploitative conditions.

Reduced emphasis on priority sector lending with financial liberalisation had led to reduction in the availability of credit to small and marginal cultivators and made cultivation more expensive. Trends in credit-deposit ratio and shares of priority sector and agriculture in total outstanding credit of commercial banks indicate that both declined in the 1990s as compared to the 1980s. Since 2001, however, there has been a turnaround, whereby there were steep increases in these ratios. Increase in rural credit since 2001 was largely due to an increase in indirect finance in agriculture and definitional changes that incorporated export-oriented and capital-intensive agriculture under priority sector lending. Main beneficiaries of this change were large agribusiness companies and big cultivators. The share of the latter in total credit outstanding and loan per account increased substantially between the mid-1990s and 2004-05.

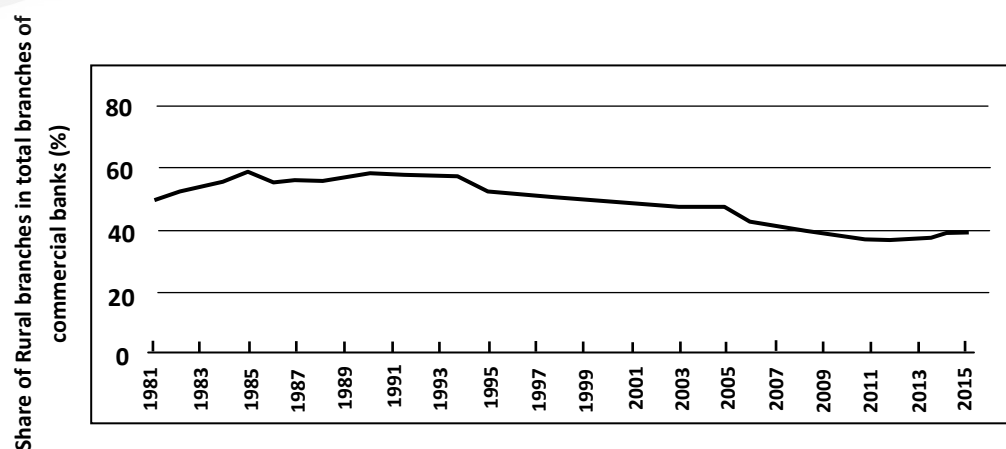


Fig. 13.2A: Trends in the declining Rural Branches in India

Source: Computed from the Handbook of Statistics, Reserve Bank of India, various years.

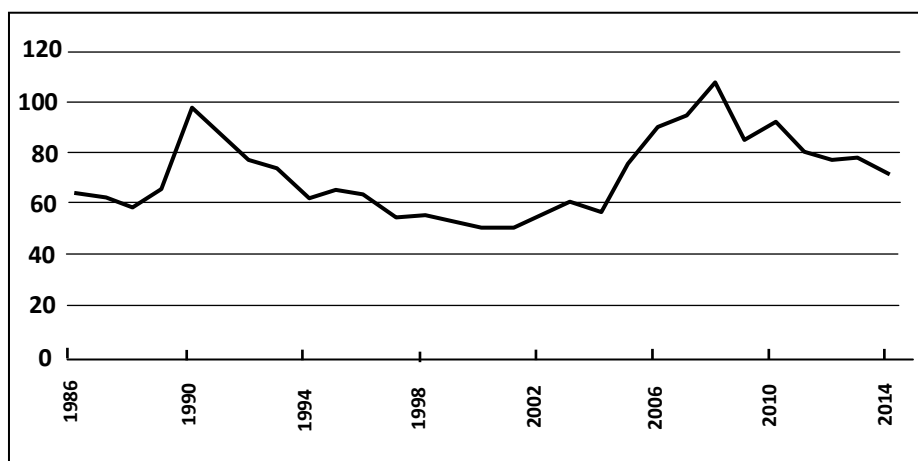


Fig. 13.2B: Trends in the declining Credit-Deposit Ratio

Source: Computed from the Handbook of Statistics, Reserve Bank of India, various years.

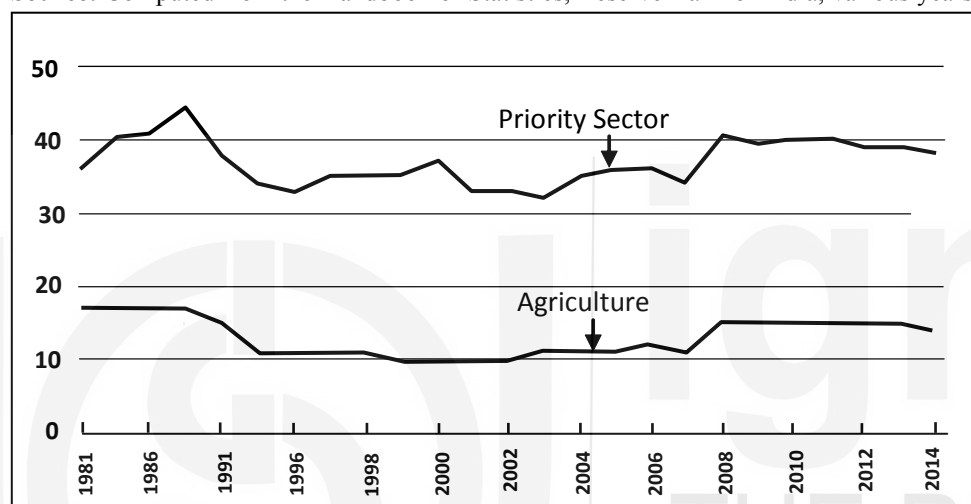


Fig. 13.2C: Shares of Priority Sector and Agriculture in Total Outstanding Credit of Commercial Banks

13.3.6 Imbalance in the use of Fertilizer

In general, the nitrogenous (N), phosphate (P) and potash (K) Fertilizers' ratio of 4:2:1 is considered to be optimum for India. The total fertilizer-use in India increased at 6 per cent per annum from 2.65 million tons of NPK in 1971-72 to 28.12 million tons in 2010-11. In the early 1970s Indian farmers applied N, P and K in the ratio of 6.0:1.9:1.0, which increased to 9.5:3.2:1 in 1992. The ratio worsened further to 10:2.9:1 in 1996. In 2012-13, the ratio of NPK use in India reached 8.2:3.2:1, which is more imbalanced compared to the early 1970s. Imbalance in the use of fertilizers in soil may also result in a loss of fertility.

13.3.7 Effectiveness of Minimum Support Prices (MSPs)

MSPs are declared prior to each sowing season (in June and October) so that farmers are aware of the minimum price the government will offer for their produce. As per a report by the NITI Aayog, a low proportion of farmers (10 per cent) were aware of MSPs before the sowing season. 62 per cent of the farmers were informed of MSPs after sowing their crops. The pricing policy of MSPs would be effective only if farmers are aware of it at the time of deciding what crops to grow. The NITI Aayog recommended that the awareness level of farmers regarding MSPs must be increased and the mediums of dissemination of this information must be strengthened.

Other issues with the implementation of the MSP regime include long distances to the procurement centres, increasing cost of transportation for farmers, irregular hours of the procurement centres, lack of covered storage godowns and inadequate storage capacity, and delays in the payment of MSPs to farmers.

13.3.8 Inadequate Availability and Access to Quality Seeds

Quality seeds are another input necessary for agricultural productivity, and good quality seeds account for 20 per cent-25 per cent of increased crop productivity. Agricultural seeds are produced by various agencies such as Indian Council of Agricultural Research and its research institutions, state agricultural universities, and national and state seeds corporations. The private sector has also started playing a role in supplying some seeds such as hybrid maize, bajra, cotton and sunflower. About 30 per cent-35 per cent of the total seeds available are produced by private and public sector companies, and farm bred seeds account for the remaining seeds. The cost of HYV varieties is too high for marginal and small farmers to afford, thus disincentivising them from purchasing these varieties. The Economic Survey 2015-16 recommended bringing in more players into the production of seeds, to improve their availability in the market and also reduce their prices.

13.3.9 Farm Mechanisation: Major Challenges

Mechanisation is another aspect with a significant impact on agricultural productivity. It makes activities such as tilling, spreading of seeds and fertilizers and harvesting more efficient, so that the cost of inputs is offset. It can also make the use of labour in agriculture more cost-effective. The status of mechanisation in agriculture varies for different activities, although the overall level of mechanisation is still less than 50 per cent, as compared to 90 per cent in developed countries.

Some challenges faced by farm mechanisation include different soil and climatic zones which require customised farm machinery, and small land holdings with lack of access to resources. To increase productivity, farm equipment which is durable, lightweight and low cost and also specific to different crops and regions should be made available for small and marginal farmers.

13.3.10 Post-harvest Activities-Inadequate Storage Facilities

After agricultural produce is harvested, it requires a robust storage infrastructure in order to minimise any losses due to adverse weather conditions or in the process of transportation. Owing to the inadequate capacity and poor conditions of storage, the quantity of food which is wasted during the harvest and post-harvest processes in the country has increased over the years. The highest losses are observed in the case of fruits and vegetables (up to 40 per cent of production), pulses (6.4 per cent-8.4 per cent) and oilseeds (5.3 per cent-9.9 per cent).

In sum, Indian agriculture continues to face challenges, some new and some old, such as: (i) declining availability/quality of soil, water and other natural resources; (ii) decreasing size of farm holdings; (iii) inefficient use of inputs and their increasing costs; (iv) scarce and more costly agriculture labour; (v) drudgery in farming operations; (vi) growing risks in farming at all its stages from pre-production to production to marketing; (vii) increasing information gap,

knowledge gap and skill gap; (viii) poor access to credit and investments; (ix) slow diffusion of relevant technologies; (x) competitiveness of quality and prices in domestic and export markets; (xi) inadequate focus on processing and value addition; (xii) inadequate rural infrastructure; (xiii) regional imbalances; (xiv) problems in retaining rural youth in agriculture; (xv) poor access to resources and services for women in agriculture; (xvi) weak institutional/sub-system linkages and convergence; and (xvii) extreme events of climate change.

Check Your Progress 2

- 1) Discuss the factors responsible for slow growth in agriculture in recent years.

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- 2) How does public investment in agriculture affects the productivity growth of the sector in India? Give reasons.

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- 3) Give a critical appraisal of the flow of institutional credit for agriculture in India.

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- 4) Do you agree with the view that shrinking farm size in India is a major factor of sluggish growth and rural poverty in India?

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13.4 TRANSFORMATION OF INDIAN AGRICULTURE: STRATEGIES FOR DEVELOPMENT (1951-2002)

Agricultural sector in India has moved from a traditional agriculture in the 1950s to the modern, technologically dynamic high capital-intensive agriculture, in

which along with food and non-food crops, horticulture and other allied activities have also expanded. In the context of Indian agriculture, three distinct phases of growth can be distinguished as follows:

- **Phase I:** Traditional Agriculture (1951-1966)
 - First sub-period (1951-61)
 - Second sub-period (1961-66)
- **Phase II:** Technologically Dynamic Agriculture with New Agriculture Strategy (NAS) (1966 - 2002)
- **Phase III:** Post Green Revolution (GR) Policy/Schematic Initiatives

13.4.1 Phase I: Traditional Agriculture (1951-1966)

This is a technologically stagnant phase in which a larger farm production becomes generally possible only through increased application of all three traditional inputs, viz. land, labour and capital. The rate of increase of output is normally smaller than the rate of increase in inputs-revealing diminishing productivity of inputs, even at a low yield. Till mid-1960s, the Indian agriculture was typically embodied within the characteristics of traditional agriculture.

The primary feature of this period was that production of agricultural crops consistently maintained an upward trend, except for small dips in two years, 1957-58 and 1959-60. Independent India implemented various programmes, listed below, as part of the effort to increase production and to ensure food security.

- Grow More Food Campaign,
- Intensive Agricultural Development Programme (1950-51),
- Community Development Programme (1952),
- National Extension Service (1953), and
- Intensive Agricultural District Programme (1960-61)

Neglect of the agricultural sector during the Second Plan, the short war with China in 1962 and with Pakistan in 1966, the widespread drought in the east India culminated in a food crisis. In 1965-66, foodgrain production in the country fell from 83 million tons in 1960-61 to 72 million tons. This led to a serious crisis in the Indian economy, as the growth in foodgrains production was inadequate to meet the consumption needs of the growing population and food imports became essential.

13.4.2 Phase II: Technologically Dynamic Agriculture with New Agriculture Strategy (NAS):1966 - 2002

This is the beginning of the process of transformation from traditional agriculture to modern agriculture. The distinguishing feature of phase II is the application of science and technology, evolved by research institutions, in a progressively large scale. The 'New Agricultural Strategy' was India's response to the grave and long - standing food crisis that culminated in the 1960s.

The salient features of the Green Revolution Strategy are briefly listed below:

- *Revolution in Biotechnology: Biotechnology* - a hybrid of genetics and chemistry determines the maximum biological performance of plants and

animals and also influences the scope and effectiveness of other forms of technology as well.

- *Green Revolution as Package Programme*: consisting of improved seeds, inorganic fertilizer, irrigation, and plant protection code (PPC) combined with agricultural R&D
- *Central Role of High Yielding Varieties or 'Miracle seeds'*: The response of the traditional varieties to fertilizer was said to be negative. When they are fertilized with chemicals, they would shoot up even faster, grow dense, stoop and lodge on the ground reducing the photosynthetic work, and in the end would yield even less.
- *An important characteristic of modern varieties is their ability to utilise nutrients and water efficiently*, and to trans-locate more of them to grain formation than to other parts such as leaf. The unique feature of these seeds is that it can grow and work uninterruptedly even under cloudy skies, which drastically cuts down the time needed to produce a crop. This results in a quantum jump in yield.
- Spread of Green Revolution limited to irrigated areas.
- *Confined to Wheat Revolution*: Presently 90 per cent of lands engaged in wheat cultivation have benefitted from this new agricultural strategy. Most of the HYV seeds are related to wheat crop and major portion of chemical fertilizers are also used in wheat cultivation.

13.4.3 Impact of Green Revolution on Production and Yield of Major Crops During the Entire Planning Period

- The rate of growth of wheat production which was 4 per cent per annum in the pre-green revolution period (1949-50 to 1964-65) shot up to 5.7 per cent per annum. During the sub-Phase I, i.e. initial phase of green revolution (1967-68 to 1980-81) the rate of productivity doubled over the period from 1.3 per cent per annum to 2.6 per cent per annum. It was due to the adoption of new HYVs of seeds in the irrigated areas in certain regions of the country.
- Other crops picked up in the decade of 1980s, for instance, rice and oilseeds did considerably better in 1980s as compared with 1967-68 to 1980-81 and green revolution started spreading to more crops and this period is termed as the maturing of green revolution.
- During pre-green revolution phase, the contribution of area growth to output growth was 50.1 per cent while that of yield growth was only 38.4 per cent. During the green revolution or post green revolution phase, yield growth became the dominant source of growth of output, i.e. 85.2 per cent of growth was accounted for by yield growth while area expansion only contributed only 14.4 per cent.
- Deceleration in agricultural growth rates in the reform period commencing in 1991: Rate of growth of food grains production fell from 2.9 per cent in 1980s to 2 per cent in 1990s and remains stagnant thereafter at around 1.2 per cent during 1999-2019. Similarly rate of growth of productivity fell from 2.7 per cent in 1980s to 1.51 per cent in 1990s and further decreases to 1.3 per cent during the period 2000-19.

Thus in the reform period, the resurgent growth in agriculture from mid-1960s was arrested.

Green revolution generated employment opportunities in diverse activities which were created as a result of multiple cropping and mechanisation of farming. It helped to stimulate non-farm economy that generated newer employment in various services such as milling, marketing, warehousing, etc. Another positive effect of Green Revolution is the ploughing back of increased profits in agriculture. Increased production and productivity of foodgrains in post green revolution period not only helped in maintaining the price stability in food grains but also strengthened the forward and backward linkages with industry.

Negative socio-economic impacts of Green Revolution

The new agriculture strategy has resulted in increased productivity and returns for farmers. However, the revolution resulted into increased income with wide interpersonal and regional inequality and inequitable asset distribution:

- Being costly, NAS was adopted by large farmers resulting in interpersonal inequalities.
- NAS was confined to only irrigated areas, resulting in an increase in regional disparities.
- Restrictive Crop Coverage: The new agriculture strategy involving use of HYV seeds was initially limited to wheat and other major crop rice responded much later. The progress of developing and application of HYV seeds in other crops especially commercial crops like oilseeds, jute, etc. has been very slow. In fact, in a certain period, a decline in the output of commercial crops is witnessed because of diversion of area under commercial crop to food crop production.

Check Your Progress 3

- 1) Give an account of major programmes implemented during first one and a half decade of planning era for enhancing the foodgrains production in India.

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- 2) Enhancing the productivity of agriculture is the key to inclusive growth which is possible only with 'strategic management' and 'technological upgradation' in the sector. Discuss.

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- 3) What are the unique characteristics of the New Agriculture Strategy (also called Green Revolution)? Do you think India needed such strategy to address its food shortage? Narrate the positive and negative impacts of the Green Revolution.

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- 4) The Green Revolution and consequent increase in agriculture produce has worked to the disadvantage of farmers. Critically analyse the statement.

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13.5 TRANSFORMATION OF INDIAN AGRICULTURE: STRATEGIES FOR DEVELOPMENT 2002-2014

The analysis of growth dynamics of agriculture after adoption of NAS reveals that agricultural development strategies must be reoriented to meet the needs of farmers and called upon the Central and State governments to evolve a strategy to rejuvenate agriculture. The new strategy requires extending the first Green Revolution to the second Green Revolution in Eastern Part of the country or to Evergreen Revolution throughout the country to ensure that growth in agriculture is sustainable in future. This called for giving more autonomy and degree of freedom to states in formulating agricultural policies.

13.5.1 Rashtriya Krishi Vikas Yojana (RKVY)

Given the fact that by the end of 1990s, rates of growth of both production and the yield plateaued or rather declined, National Development Council (NDC) resolved on 29.5.2007 that agricultural development strategies must be reoriented to meet the needs of farmers and called upon the Central and State governments to evolve a strategy to rejuvenate agriculture. In compliance of the NDC resolution Ministry of Agriculture and Farmers Welfare launched a new Scheme named Rashtriya Krishi Vikas Yojana (RKVY) in 2007 with the following objectives:

- To incentivise the states that increase their investment in agriculture and allied sectors;
- To provide flexibility and autonomy to the States in planning and executing programmes for agriculture;
- To ensure the preparation of agriculture plans for the districts and states;

- To achieve the goal of reducing the yield gaps in important crops;
- To maximise returns to the farmers;
- To address the agriculture and allied sectors in an integrated manner.

The macro-level impact of RKVY at both the national and state level has been measured using three growth indicators. These indicators are: (i) growth in GDP, overall GSDP and agricultural GSDP (ii) growth in agricultural area and production and (iii) percentage of agriculture expenditure out of states' budget and agriculture GSDP. The growth in agricultural GDP of the country was 2.4 per cent per annum in the 10th Plan that increased to 3.7 per cent per annum in the 11th Plan. The overall growth rate of GSDP of 28 states (excluding the UTs) also went-up from 7.61 per cent in 10th FYP to 8.63 per cent in the 11th FYP. Similarly, the overall agriculture GSDP of 28 states grew from 2.38 per cent in 10th FYP to 3.66 per cent in 11th FYP.

13.5.2 Second Green Resolution

Green Revolution accelerated the yields of major food crops such as paddy, wheat, millets and oil seeds, particularly in the states of Punjab, Haryana, parts of Uttar Pradesh and Rajasthan. Green Revolution in India created a huge impact in terms of crop production but it had several negative impacts like gradual loss of soil fertility, increasing alkalinity and salinity, water-logging, depletion of ground-water resources, decreasing bio-diversity, chemical poisoning of soils, surface water and food. The first Green Revolution was launched to ensure food security as there was severe scarcity of food in the country. Now our food supply is well secure and meeting the growing needs is within reach. Therefore, the second Green Revolution should aim at promoting sustainable livelihood, enabling the poor to come out of poverty by generating gainful self-employment. There is a need for a second green revolution which could potentially address the negative impacts of the climate change through some kinds of farming like:

- *Agro-forestry on Degraded Lands:* The focus should be on sustainable use of degraded and low fertility lands deprived of irrigation. Such lands can be profitably used for establishment of drought tolerant fruit crops and agriculture-horticulture pastures.
- *Farming on Wastelands:* Apart from un-irrigated lands, there are large stretches of wastelands in India. Among the estimated 40-50 million ha of wastelands, more than 9 million ha are sodic wastelands (Fertile irrigated fields which turned sodic due to excessive use of water for irrigation and poor drainage facilities). With reclamation of sodic lands, it is possible to enhance food production while creating employment for 8-10 million people.
- *Organic Agriculture:* cultivation without any use of chemical inputs like mineral fertilizers and chemical pesticides.
- *Green Agriculture:* cultivation with the help of integrated pest management, integrated nutrient supply and integrated natural resource management systems.
- *Eco-agriculture:* Based on conservation of soil, water and biodiversity and the application of traditional knowledge and ecological prudence.
- *EM Agriculture:* system of farming using effective microorganisms (EM).